



Top 59 CAT Inequalities Questions With Video Solutions

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Questions

Instructions

For the following questions answer them individually

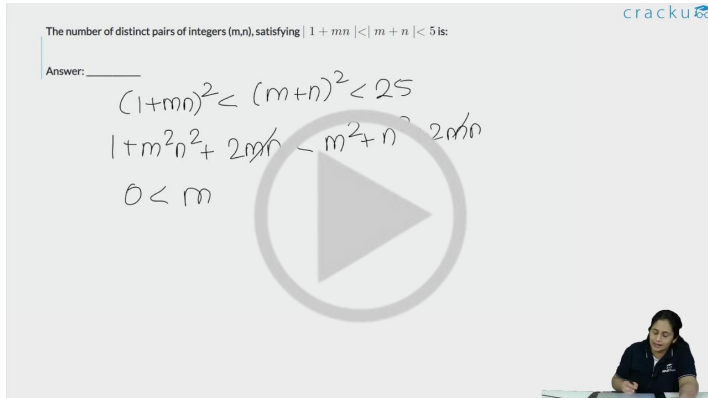
Question 1

The number of distinct pairs of integers (m,n) , satisfying $|1 + mn| < |m + n| < 5$ is:

The number of distinct pairs of integers (m,n) , satisfying $|1 + mn| < |m + n| < 5$ is:

Answer: _____

$(1+mn)^2 < (m+n)^2 < 25$
 $1+m^2n^2+2mn < m^2+n^2+2mn$
 $0 < n$



 VIDEO SOLUTION

Question 2

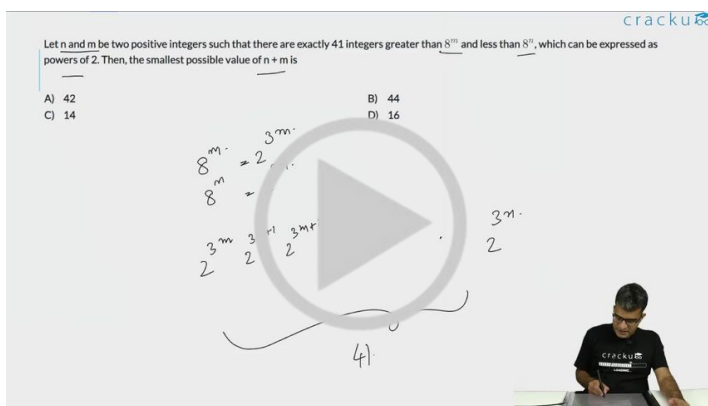
Let n and m be two positive integers such that there are exactly 41 integers greater than 8^m and less than 8^n , which can be expressed as powers of 2. Then, the smallest possible value of $n + m$ is

- A 42
- B 44
- C 14
- D 16

Let n and m be two positive integers such that there are exactly 41 integers greater than 8^m and less than 8^n , which can be expressed as powers of 2. Then, the smallest possible value of $n + m$ is

A) 42
B) 44
C) 14
D) 16

$8^m = 2^{3m}$
 $8^n = 2^{3n}$
 $2^{3m} < 2^{3m+1} < 2^{3m+2} < \dots < 2^{3n-1} < 2^{3n}$
The number of integers between 2^{3m} and 2^{3n} is $2^{3n} - 2^{3m} - 1$. This must equal 41.



 VIDEO SOLUTION

Question 3

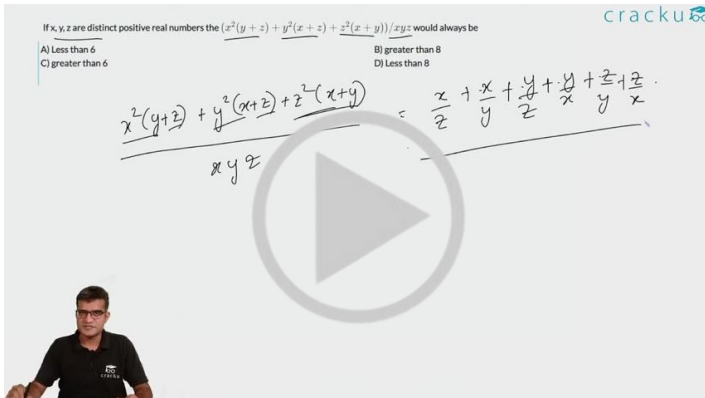
If x, y, z are distinct positive real numbers the $(x^2(y+z) + y^2(x+z) + z^2(x+y))/xyz$ would always be

- A Less than 6

- B greater than 8
- C greater than 6
- D Less than 8

If x, y, z are distinct positive real numbers the $\frac{x^2(y+z) + y^2(x+z) + z^2(x+y)}{xyz}$ would always be

A) Less than 6
B) greater than 8
C) greater than 6
D) Less than 8

$$\frac{x^2(y+z) + y^2(x+z) + z^2(x+y)}{xyz} = \frac{x}{z} + \frac{x}{y} + \frac{y}{z} + \frac{y}{x} + \frac{z}{y} + \frac{z}{x}$$


VIDEO SOLUTION

CAT Mock Test

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Question 4

Let a, b, c, d be four integers such that $a+b+c+d = 4m+1$ where m is a positive integer. Given m , which one of the following is necessarily true?

- A The minimum possible value of $a^2 + b^2 + c^2 + d^2$ is $4m^2 - 2m + 1$
- B The minimum possible value of $a^2 + b^2 + c^2 + d^2$ is $4m^2 + 2m + 1$
- C The maximum possible value of $a^2 + b^2 + c^2 + d^2$ is $4m^2 - 2m + 1$
- D The maximum possible value of $a^2 + b^2 + c^2 + d^2$ is $4m^2 + 2m + 1$

VIEW SOLUTION

Question 5

The number of integer solutions of the equation $(x^2 - 10)^{(x^2 - 3x - 10)} = 1$ is

The number of integer solutions of the equation $(x^2 - 10)(x^2 - 3x - 10) = 1$ is

Answer: _____

$A > 1$
 $x^2 - 10 = 1$
 $x^2 = 11$
 $x = \pm \sqrt{11}$

$B = 0$ (A=0)
 $x^2 - 3x - 10 = 0$

$A = -1$ & B is even.

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Question 6

The number of distinct integer solutions (x, y) of the equation $|x + y| + |x - y| = 2$, is

The number of distinct integer solutions (x, y) of the equation $|x + y| + |x - y| = 2$, is

Answer: _____

One! 2 Solutions
 $x = y$
 $|2x| = 2$
 $|x| = 1$
 $x = \pm 1$

$x + y$
 $x - y \rightarrow \text{integers}$
 $0, 2$
 $x = y$
 $-y = -x$

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CAT Online Coaching

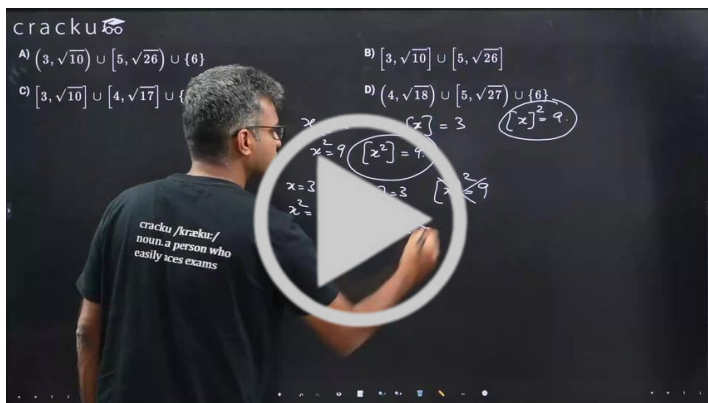
your road to IIM stars here

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Question 7

Let $3 \leq x \leq 6$ and $[x^2] = [x]^2$, where $[x]$ is the greatest integer not exceeding x . If set S represents all feasible values of x , then a possible subset of S is

- A $(3, \sqrt{10}) \cup [5, \sqrt{26}) \cup \{6\}$
- B $[3, \sqrt{10}] \cup [5, \sqrt{26}]$
- C $[3, \sqrt{10}] \cup [4, \sqrt{17}] \cup \{6\}$
- D $(4, \sqrt{18}) \cup [5, \sqrt{27}) \cup \{6\}$

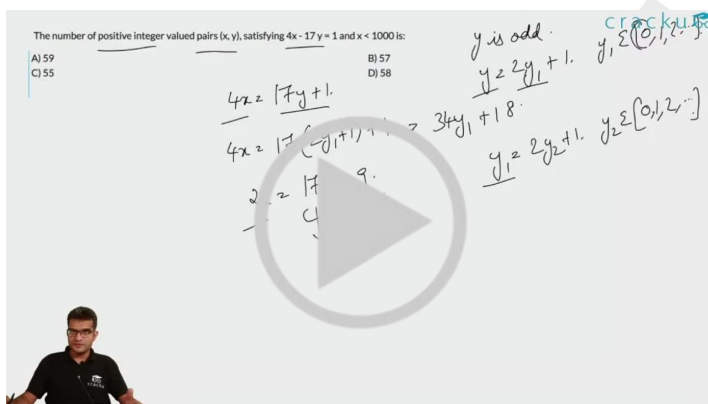


VIDEO SOLUTION

Question 8

The number of positive integer valued pairs (x, y) , satisfying $4x - 17y = 1$ and $x < 1000$ is:

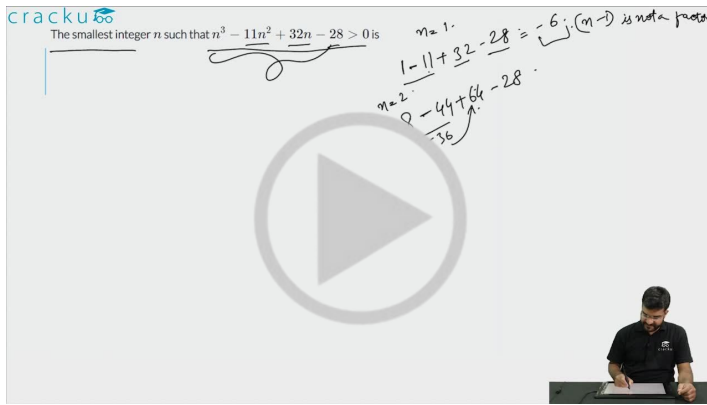
- A 59
- B 57
- C 55
- D 58



VIDEO SOLUTION

Question 9

The smallest integer n such that $n^3 - 11n^2 + 32n - 28 > 0$ is

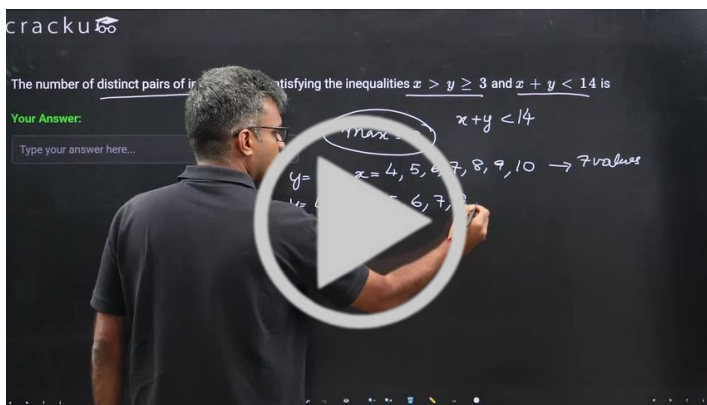


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Question 10

The number of distinct pairs of integers (x, y) satisfying the inequalities $x > y \geq 3$ and $x + y < 14$ is



VIDEO SOLUTION

Question 11

From any two numbers x and y , we define $x * y = x + 0.5y - xy$. Suppose that both x and y are greater than 0.5. Then

$x * x < y * y$ if

- A $1 > x > y$
- B $x > 1 > y$
- C $1 > y > x$
- D $y > 1 > x$

VIDEO SOLUTION

Question 12

If x and y are real numbers such that $x^2 + (x - 2y - 1)^2 = -4y(x + y)$, then the value $x - 2y$ is

- A 0
- B 1
- C -1
- D 2



[VIDEO SOLUTION](#)



CAT Syllabus PDF



Question 13

The number of integer solutions of equation $2|x|(x^2 + 1) = 5x^2$ is

[VIDEO SOLUTION](#)

Question 14

In an examination, the average marks of 4 girls and 6 boys is 24. Each of the girls has the same marks while each of the boys has the same marks. If the marks of any girl is at most double the marks of any boy, but not less than the marks of any boy, then the number of possible distinct integer values of the total marks of 2 girls and 6 boys is

- A 21
- B 20
- C 22
- D 19

A) 21
C) 22

Any non-zero real numbers x, y such that $y \neq 3$ and $\frac{x}{y} < \frac{x+3}{y-3}$, Will satisfy the condition.

- A** $\frac{x}{y} < \frac{y}{x}$
- B** If $y < 0$, and $-x < y$
- C** If $y > 10$, and $-x > y$
- D** If $x < 0$, and $-x < y$

A) $\frac{x}{y} < \frac{z}{y}$
Q) If $y > 10$, and $-x > y$



CAT Formula PDF


If a certain amount of money is divided equally among n persons, each one receives Rs 352. However, if two persons receive Rs 506 each and the remaining amount is divided equally among the other persons, each of them receive less than or equal to Rs 330. Then, the maximum possible value of n is

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If a certain amount of money is divided equally among n persons, each one receives Rs 352. However, if two persons receive Rs 506 each and the remaining amount is divided equally among the other persons, each of them receive less than or equal to Rs 330. Then, the maximum possible value of n is

Answer: _____

Handwritten notes:

$$\begin{aligned} \text{Total money} &= 352n \\ \text{Remaining Money} &= 352n - 2 \times 506 \\ &= 352n - 1012 \end{aligned}$$


VIDEO SOLUTION

Question 17

If $p^2 + q^2 - 29 = 2pq - 20 = 52 - 2pq$, then the difference between the maximum and minimum possible value of $(p^3 - q^3)$

- A 243
- B 486
- C 378
- D 189

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If $p^2 + q^2 - 29 = 2pq - 20 = 52 - 2pq$, then the difference between the maximum and minimum possible value of $(p^3 - q^3)$

A) 243
C) 378

B) 486
D) 189

Handwritten notes:

$$\begin{aligned} 4pq &= 42 \\ pq &= 10.5 \\ p^2 - 2pq + q^2 &= 9 \\ (p-q)^2 &= 9 \\ p-q &= 3 \\ p-q &= -3 \end{aligned}$$

$$(p-q)(p^2 + pq + q^2)$$


VIDEO SOLUTION

Question 18

In the XY-plane, the area, in sq. units, of the region defined by the inequalities $y \geq x + 4$ and $-4 \leq x^2 + y^2 + 4(x - y) \leq 0$ is

- A 2π
- B 4π
- C π
- D 3π

In the XY-plane, the area, in sq. units, of the region defined by the inequalities $y \geq x + 4$ and $-4 \leq x^2 + y^2 + 4(x - y) \leq 0$ is

A) 2π
B) 4π
C) π
D) 3π

Handwritten solution:

$$y \geq x + 4$$

$$x^2 + y^2 + 4(x - y) - 8 \leq 0$$

$$x^2 + 4x + y^2 - 4y - 8 + 4 = 0$$

$$(x+2)^2 + (y-2)^2 - 8 = 0$$

$$(x+2)^2 + (y-2)^2 \leq 8$$

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Question 19

If $x > 5$ and $y < -1$, then which of the following statements is true?

- A $(x + 4y) > 1$
- B $x > -4y$
- C $-4x < 5y$
- D None of these

If $x > 5$ and $y < -1$, then which of the following statements is true?

A) $(x + 4y) > 1$
B) $x > -4y$
C) $-4x < 5y$
D) None of these

Handwritten solution:

$$x > 5 \quad y < -1$$

$$x = 6 \quad y = -100$$

$$x + 4y = 6 - 400 = -394$$

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VIDEO SOLUTION

Question 20

x and y are real numbers satisfying the conditions $2 < x < 3$ and $-8 < y < -7$. Which of the following expressions will have the least value?

- A x^2y
- B xy^2
- C $5xy$

D None of these

[VIEW SOLUTION](#)

Question 21

Let p, q and r be three natural numbers such that their sum is 900, and r is a perfect square whose value lies between 150 and 500. If p is not less than $0.3q$ and not more than $0.7q$, then the sum of the maximum and minimum possible values of p is

[VIEW SOLUTION](#)

CAT Percentile Predictor



Question 22

The set of all real values of x for which $(x^2 - |x + 9| + x) > 0$, is

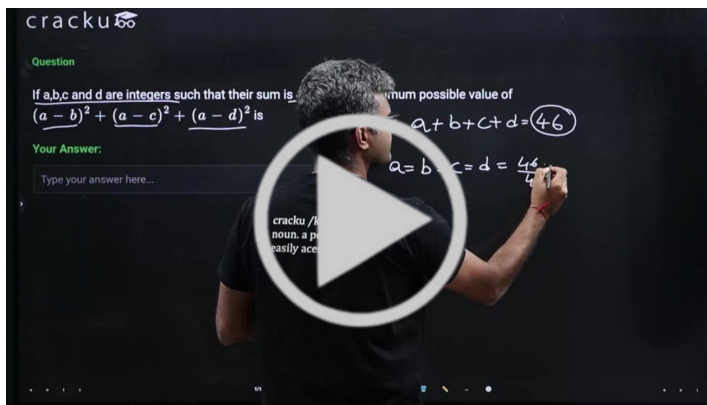
- A $(-\infty, -3) \cup (3, \infty)$
- B $(-\infty, -9) \cup (3, \infty)$
- C $(-9, -3) \cup (3, \infty)$
- D $(-\infty, -9) \cup (9, \infty)$



[VIDEO SOLUTION](#)

Question 23

If a, b, c and d are integers such that their sum is 46, then the minimum possible value of $(a - b)^2 + (a - c)^2 + (a - d)^2$ is

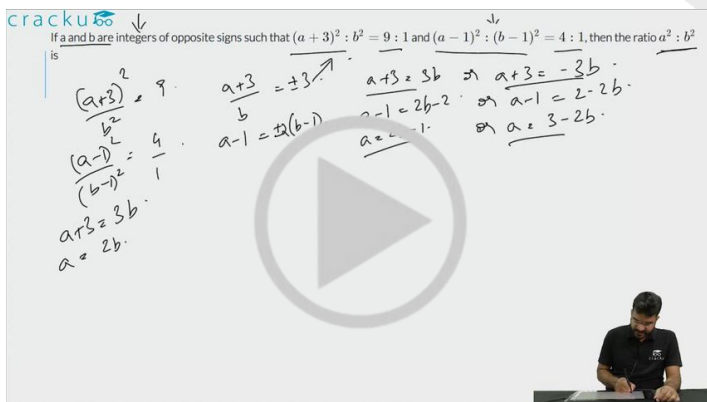


VIDEO SOLUTION

Question 24

If a and b are integers of opposite signs such that $(a + 3)^2 : b^2 = 9 : 1$ and $(a - 1)^2 : (b - 1)^2 = 4 : 1$, then the ratio $a^2 : b^2$ is

- A 9:4
- B 81:4
- C 1:4
- D 25:4



VIDEO SOLUTION



Question 25

$f(x) = \frac{x^2 + 2x - 15}{x^2 - 7x - 18}$ is negative if and only if

- A $-5 < x < -2$ or $3 < x < 9$
- B $x < -5$ or $-2 < x < 3$
- C $-2 < x < 3$ or $x > 9$

D $x < -5$ or $3 < x < 9$

$f(x) = \frac{x^2+2x-15}{x^2+2x-18}$ is negative if and only if

A) $-5 < x < -2$ or $3 < x < 9$
 B) $x < -5$ or $-2 < x < 3$
 C) $-2 < x < 3$ or $x > 9$
 D) $x < -5$ or $3 < x < 9$

$x^2+2x-15 = (x+5)(x-3)$
 $x^2+2x-18 = (x-9)(x+2)$

$\frac{(x+5)(x-3)}{(x+2)(x-9)} < 0$

$\frac{(x+5)(x-3)(x+2)(x-9)}{-9 \quad 3 \quad -2 \quad 9} < 0$

VIDEO SOLUTION

Question 26

The number of distinct integer values of n satisfying $\frac{4-\log_2 n}{3-\log_4 n} < 0$, is

The number of distinct integer values of n satisfying $\frac{4-\log_2 n}{3-\log_4 n} < 0$, is

Answer: _____

$\frac{4-\log_2 n}{3-\log_4 n} < 0$

$\log_4 n = p$
 $n = \log_2^2 n = \frac{1}{2} \log_2 n$
 $\log_2 n = 2p$

$\frac{4-2p}{3-p} < 0$
 $\frac{2-p}{3-p} < 0$
 $(2-p)(3-p) < 0$

VIDEO SOLUTION

Question 27

The number of solutions of the equation $2x + y = 40$ where both x and y are positive integers and $x \leq y$ is:

- A 7
- B 13
- C 14
- D 18
- E 20

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The number of solutions of the equation $2x + y = 40$ where both x and y are positive integers and $x \leq y$ is:

A) 7
B) 13
C) 14
D) 20
E) 20

Handwritten solution:

$$2x + y = 40$$

1	38
2	36
3	34
4	32
5	30
...	...
17	2

Video player interface with a play button and a person in the bottom left corner.

VIDEO SOLUTION

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Live Classes by IIMA alumni

Image of two people and a book icon.

Question 28

If $x^2 + 5y^2 + z^2 = 2y(2x + z)$, then which of the following statements is(are) necessarily true?

A. $x = 2y$ B. $x = 2z$ C. $2x = z$

- A Only A
B B and C
C A and B
D None of these

VIEW SOLUTION

Question 29

Let n be any natural number such that $5^{n-1} < 3^{n+1}$. Then, the least integer value of m that satisfies $3^{n+1} < 2^{n+m}$ for each such n , is

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Let n be any natural number such that $5^{n-1} < 3^{n+1}$. Then, the least integer value of m that satisfies $3^{n+1} < 2^{n+m}$ for each such n , is

Answer: _____

Handwritten solution:

$5^n < 3^{n+1}$

$\left(\frac{5}{3}\right)^n < 3$

$n < 15$

$n = 1, 2, 3, 4, 5$

$3^{n+1} < 2^{n+m}$

Video player interface with a play button and a person in the bottom right corner.

VIDEO SOLUTION

Question 30

The population of a town in 2020 was 100000. The population decreased by $y\%$ from the year 2020 to 2021, and increased by $x\%$ from the year 2021 to 2022, where x and y are two natural numbers. If population in 2022 was greater than the population in 2020 and the difference between x and y is 10, then the lowest possible population of the town in 2021 was

- A 72000
- B 74000
- C 73000
- D 75000

VIDEO SOLUTION

Question 31

If $pqr = 1$, the value of the expression $1/(1 + p + q^{-1}) + 1/(1 + q + r^{-1}) + 1/(1 + r + p^{-1})$

- A $p+q+r$
- B $1/(p+q+r)$
- C 1
- D $p^{-1} + q^{-1} + r^{-1}$

Question 32

Which of the following values of x do not satisfy the inequality $(x^2 - 3x + 2 > 0)$ at all?

- A $1 \leq x \leq 2$
- B $-1 \geq x \geq -2$
- C $0 \leq x \leq 2$
- D $0 \geq x \geq -2$

Which of the following values of x do not satisfy the inequality $(x^2 - 3x + 2 > 0)$ at all?

A) $1 \leq x \leq 2$ B) $-1 \geq x \geq -2$
 C) $0 \leq x \leq 2$ D) $0 \geq x \geq -2$

$x^2 - 3x + 2 \leq 0$
 $(x-1)(x-2) \leq 0$

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Question 33

For a real number x the condition $|3x - 20| + |3x - 40| = 20$ necessarily holds if

- A $10 < x < 15$
- B $9 < x < 14$
- C $7 < x < 12$
- D $6 < x < 11$

For a real number x the condition $|3x - 20| + |3x - 40| = 20$ necessarily holds if

A) $10 < x < 15$ B) $9 < x < 14$
 C) $7 < x < 12$ D) $6 < x < 11$

$3x < 20$
 $20 - 3x + 40 - 3x = 20$
 $60 - 6x = 20$
 $40 = 6x$
 $x = \frac{20}{3}$

$20 \leq 3x \leq 40$
 $\frac{20}{3} \leq x \leq \frac{40}{3}$

$3x > 40$
 $3x - 20 + 3x - 40 = 20$
 $6x - 60 = 20$
 $6x = 80$
 $x = \frac{40}{3}$

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Step by step, heading towards 99+ percentile



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m is the smallest positive integer such that for any integer $n \geq m$, the quantity $n^3 - 7n^2 + 11n - 5$ is positive. What is the value of m ?

A) 4
B) 5
C) 6

$n = 1, (m-1)$
 $1 - 7 + 11 - 5 = 0.$

$$\frac{n^3 - 7n^2 + 11n - 5}{n^3 - n^2 - 6n^2 + 11n - 5}$$

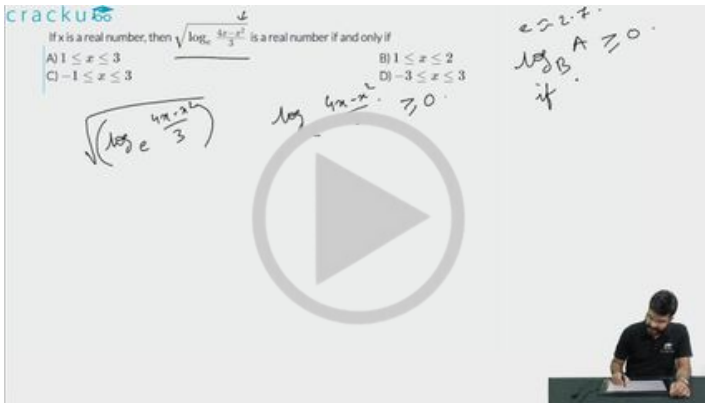
$$= \frac{6n(n-1)}{n^2(n-1)}$$

VIDEO SOLUTION

Question 36

If x is a real number, then $\sqrt{\log_e \frac{4x-x^2}{3}}$ is a real number if and only if

- A $1 \leq x \leq 3$
- B $1 \leq x \leq 2$
- C $-1 \leq x \leq 3$
- D $-3 \leq x \leq 3$



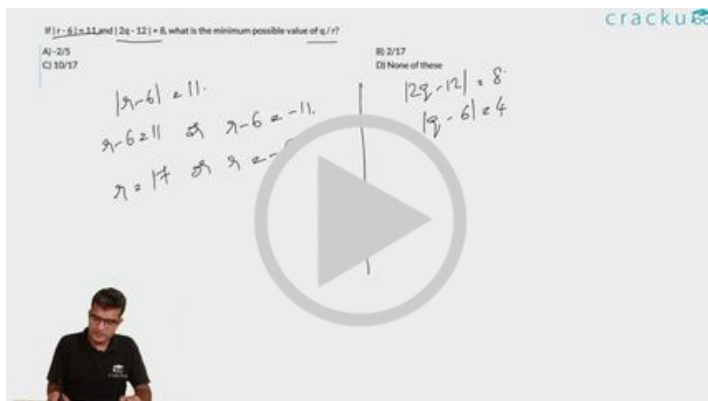
VIDEO SOLUTION



Question 37

If $|r - 6| = 11$ and $|2q - 12| = 8$, what is the minimum possible value of q / r ?

- A $-2/5$
- B $2/17$
- C $10/17$
- D None of these

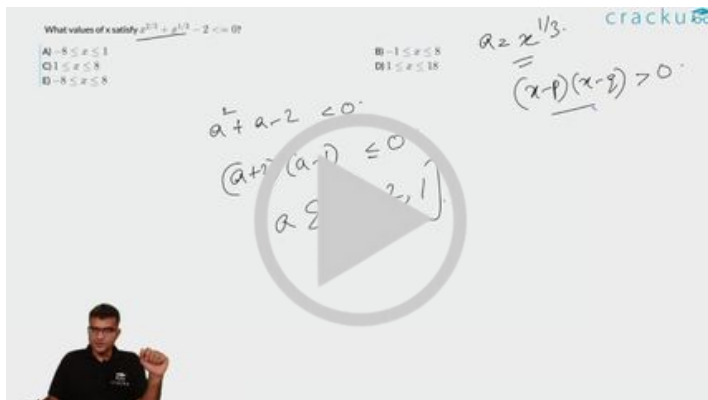


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Question 38

What values of x satisfy $x^{2/3} + x^{1/3} - 2 \leq 0$?

- A $-8 \leq x \leq 1$
- B $-1 \leq x \leq 8$
- C $1 \leq x \leq 8$
- D $1 \leq x \leq 18$
- E $-8 \leq x \leq 8$



VIDEO SOLUTION

Question 39

If $x > 2$ and $y > -1$, then which of the following statements is necessarily true?

- A $xy > -2$
- B $-x < 2y$
- C $xy < -2$
- D $-x > 2y$

If $x > 2$ and $y > -1$, then which of the following statements is necessarily true?

A) $xy > 2$
 B) $-x < 2y$
 C) $xy < 2$
 D) $-x > 2y$

Handwritten notes: $x > 2$, $y > -1$, $x \approx 1000$, $y \approx -900$. A number line shows x and y with arrows indicating their ranges.

VIDEO SOLUTION

CAT Study Material

All CAT Free Resources

Question 40

If x , y and z are real numbers such that $x + y + z = 5$ and $xy + yz + zx = 3$, what is the largest value that x can have?

- A $5/3$
- B $\sqrt{19}$
- C $13/3$
- D None of these

If x , y and z are real numbers such that $x + y + z = 5$ and $xy + yz + zx = 3$, what is the largest value that x can have?

A) $5/3$
 B) $\sqrt{19}$
 C) $13/3$
 D) None of these

Handwritten notes: $z = 5 - x - y$, $xy + 5y - xy - y^2 + 5x - x^2 - xy = 3$, $xy - 4 + x^2 - 5y + 3 = 0$, $y^2 + 1(x-5)^2 - 5x + 3 = 0$.

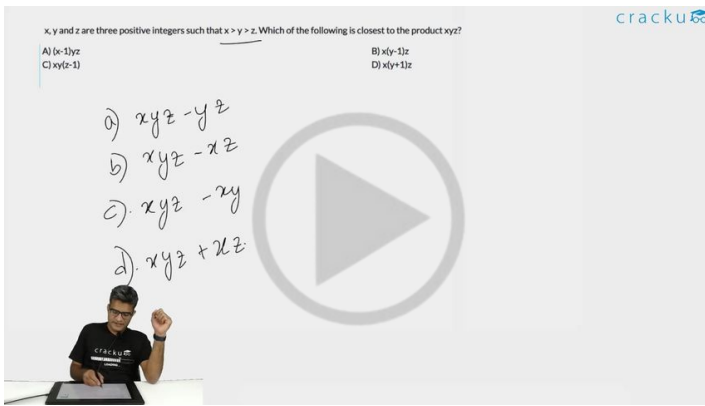
VIDEO SOLUTION

Question 41

x , y and z are three positive integers such that $x > y > z$. Which of the following is closest to the product xyz ?

- A $(x-1)yz$
- B $x(y-1)z$
- C $xy(z-1)$

D $x(y+1)z$



VIDEO SOLUTION

Question 42

Let x and y be two positive numbers such that $x + y = 1$.

Then the minimum value of $(x + \frac{1}{x})^2 + (y + \frac{1}{y})^2$ is

- A 12
- B 20
- C 12.5
- D 13.3



VIDEO SOLUTION



Question 43

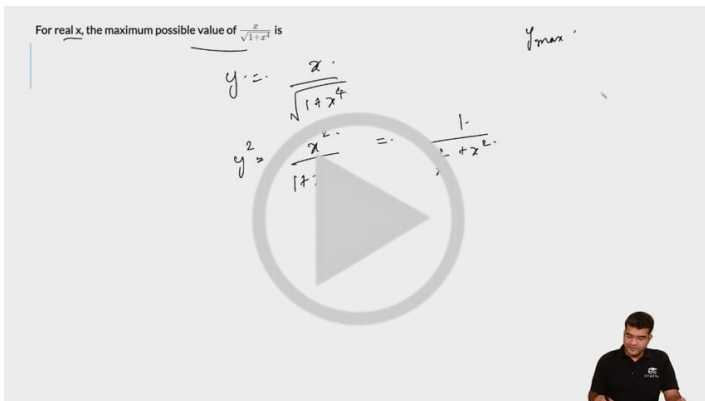
For real x , the maximum possible value of $\frac{x}{\sqrt{1+x^4}}$ is

- A $\frac{1}{2}$

B 1

C $\frac{1}{\sqrt{3}}$

D $\frac{1}{\sqrt{2}}$



VIDEO SOLUTION

Question 44

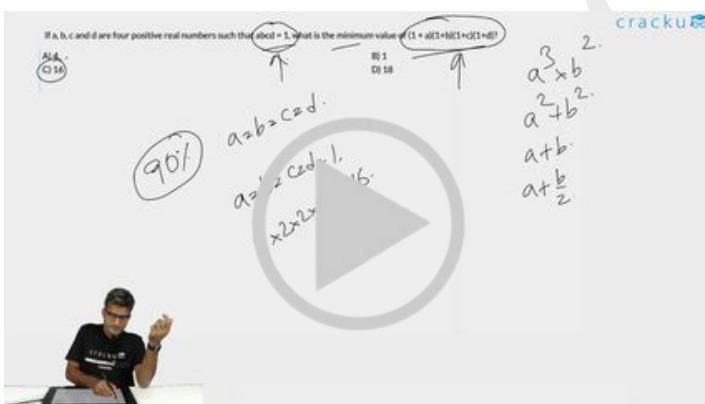
If a, b, c and d are four positive real numbers such that $abcd = 1$, what is the minimum value of $(1+a)(1+b)(1+c)(1+d)$?

A 4

B 1

C 16

D 18



VIDEO SOLUTION

Question 45

Among 100 students, x_1 have birthdays in January, x_2 have birthdays in February, and so on. If $x_0 = \max(x_1, x_2, \dots, x_{12})$, then the smallest possible value of x_0 is

A 8

B 9

C 10

D 12

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Among 100 students, x_1 have birthdays in January, x_2 have birthdays in February, and so on. If $x_0 = \max(x_1, x_2, \dots, x_{12})$, then the smallest possible value of x_0 is

A) 8
B) 9
C) 10
D) 12

$x_1 + x_2 + \dots + x_{12} = 100$

$x_0 = 100$

$x_0 =$

0 + ... + 100
0 50



 VIDEO SOLUTION

100+ CAT Quant Questions

Question 46

For how many integers n , will the inequality $(n - 5)(n - 10) - 3(n - 2) \leq 0$ be satisfied?

[illegible]

 VIDEO SOLUTION

Question 47

if x and y are positive real numbers satisfying $x + y = 102$, then the minimum possible value of $2601\left(1 + \frac{1}{x}\right)\left(1 + \frac{1}{y}\right)$ is

If x and y are positive real numbers satisfying $x + y = 102$, then the minimum possible value of $2601\left(1 + \frac{1}{x}\right)\left(1 + \frac{1}{y}\right)$ is

$2601 \left(1 + \frac{1}{x}\right) \left(1 + \frac{1}{y}\right)$

$1 + \frac{1}{x} + \frac{1}{y} + \frac{1}{xy}$

$\frac{x+y+1}{xy}$

$\frac{103}{xy}$ min

x




VIDEO SOLUTION

Question 48

The number of pairs of integers (x, y) satisfying $x \geq y \geq -20$ and $2x + 5y = 99$

The number of pairs of integers (x, y) satisfying $x \geq y \geq -20$ and $2x + 5y = 99$

$2x + 5y = 99$

$2x \leq 2x + 5y$

$2x \leq 99$ $\therefore x \leq 49$

$2x \geq 99$ $\therefore x \geq 49.5$

$x \geq 50$




VIDEO SOLUTION

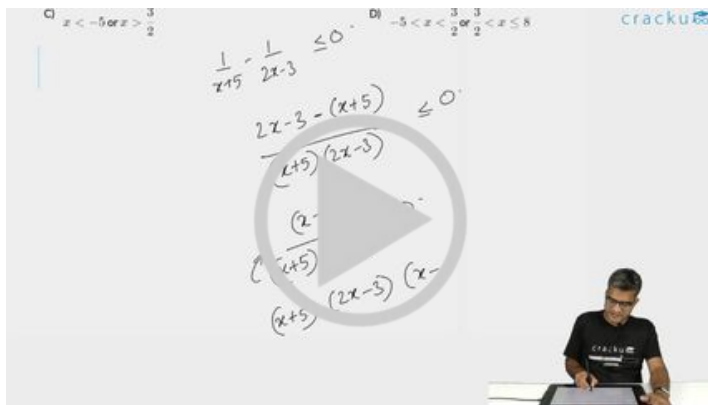
100+ CAT DILR Questions



Question 49

All the values of x satisfying the inequality $\frac{1}{x+5} \leq \frac{1}{2x-3}$ are

- A $x < -5$ or $\frac{3}{2} < x \leq 8$
- B $-5 < x < \frac{3}{2}$ or $x > \frac{3}{2}$
- C $x < -5$ or $x > \frac{3}{2}$
- D $-5 < x < \frac{3}{2}$ or $\frac{3}{2} < x \leq 8$

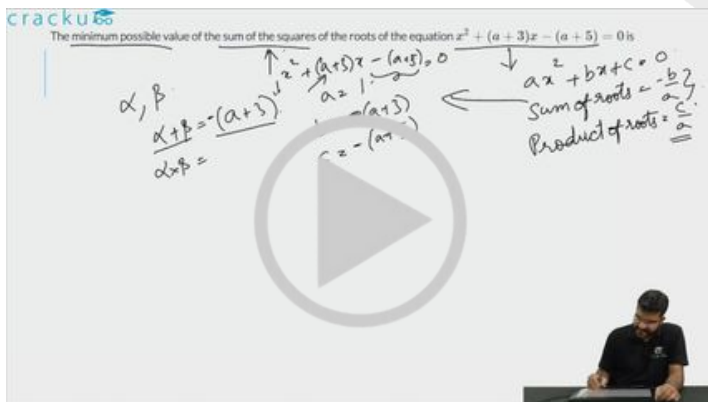


VIDEO SOLUTION

Question 50

The minimum possible value of the sum of the squares of the roots of the equation $x^2 + (a+3)x - (a+5) = 0$ is

- A 1
- B 2
- C 3
- D 4



VIDEO SOLUTION

Question 51

The number of integers n satisfying $-n+2 \geq 0$ and $2n \geq 4$ is

- A 0
- B 1
- C 2
- D 3

The number of integers n satisfying $-n+2 \geq 0$ and $2n \geq 4$ is

A) 0
C) 2

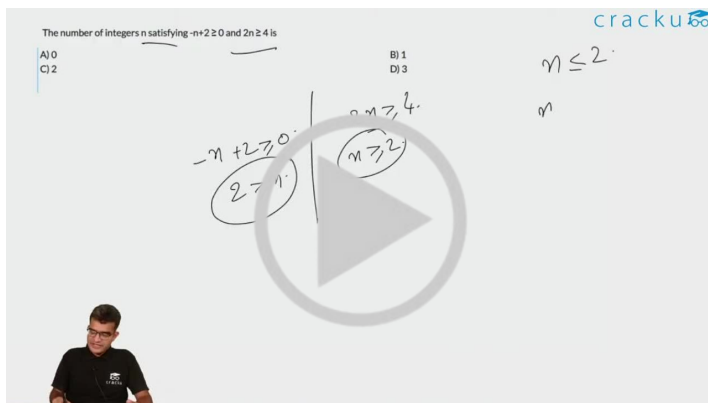
B) 1
D) 3

$n \leq 2$

$-n+2 \geq 0$
 $2 \geq n$

$2n \geq 4$
 $n \geq 2$

$n = 2$

VIDEO SOLUTION

100+ CAT VARC Questions



Question 52

If x and y satisfy the equations $|x| + x + y = 15$ and $x + |y| - y = 20$, then $(x - y)$ equals

- A 20
- B 15
- C 5
- D 10

If x and y satisfy the equations $|x| + x + y = 15$ and $x + |y| - y = 20$, then $(x - y)$ equals

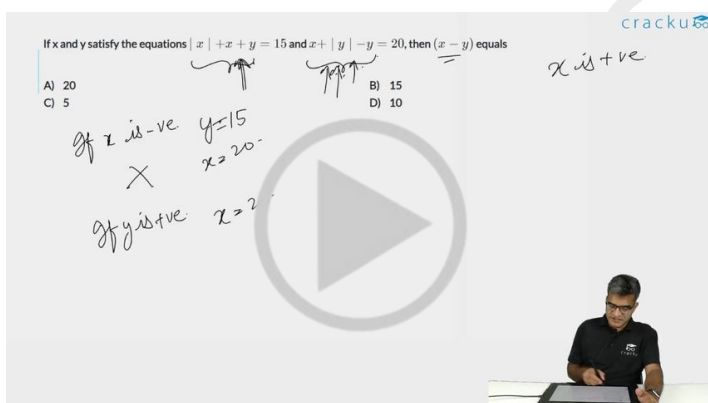
A) 20
C) 5

B) 15
D) 10

x is +ve

if x is -ve $y = 15$
 $x = 20$

if y is +ve $x = 20$

VIDEO SOLUTION

Question 53

The number of integers n that satisfy the inequalities $|n - 60| < |n - 100| < |n - 20|$ is

- A 21
- B 19
- C 18

D 20

The number of integers n that satisfy the inequalities $|n - 60| < |n - 100| < |n - 20|$ is

A) 21
B) 19
C) 18
D) 20

Handwritten notes on the screen show the solution process, including the inequalities $n < 20$, $60 - n < 100 - n$, and $100 - n < 20 - n$.



VIDEO SOLUTION

Question 54

If n is a positive integer such that $(\sqrt[7]{10})(\sqrt[7]{10})^2 \dots (\sqrt[7]{10})^n > 999$, then the smallest value of n is

If n is a positive integer such that $(\sqrt[7]{10})(\sqrt[7]{10})^2 \dots (\sqrt[7]{10})^n > 999$, then the smallest value of n is

Answer: _____

Handwritten notes on the screen show the solution process, including the inequality $10^{1/n} > 999$ and the calculation $10^{1/7} \approx 1.33$.



VIDEO SOLUTION

CAT Expected Questions PDF



Question 55

If u, v, w and m are natural numbers such that $u^m + v^m = w^m$, then which one of the following is true?

- A $m \geq \min(u, v, w)$
- B $m \geq \max(u, v, w)$
- C $m < \min(u, v, w)$
- D None of these

VIEW SOLUTION

Question 56

Each family in a locality has at most two adults, and no family has fewer than 3 children.

Considering all the families together, there are more adults than boys, more boys than girls, and more girls than families.

Then the minimum possible number of families in the locality is

- A 4
- B 5
- C 2
- D 3

Each family in a locality has at most two adults, and no family has fewer than 3 children.
Considering all the families together, there are more adults than boys, more boys than girls, and more girls than families.
Then the minimum possible number of families in the locality is

A) 4
B) 5
C) 2

Handwritten notes on a screen:

1 Family 2A 1B 0G. X.
2 Family 4 3B 2G. X.

cracku

VIDEO SOLUTION

Instructions

For these questions the following functions have been defined.

$$la(x, y, z) = \min(x + y, y + z)$$

$$le(x, y, z) = \max(x - y, y - z)$$

$$ma(x, y, z) = \frac{1}{2}(le(x, y, z) + la(x, y, z))$$

Question 57

For $x=15$, $y=10$ and $z=9$, find the value of $le(x, \min(y, x-z), le(9, 8, ma(x, y, z)))$.

- A 5
- B 12
- C 9
- D 4

VIEW SOLUTION

cracku

CAT Important Topics

Question 58

Given that $x > y > z > 0$. Which of the following is necessarily true?

- A $la(x, y, z) < le(x, y, z)$
- B $ma(x, y, z) < la(x, y, z)$
- C $ma(x, y, z) < le(x, y, z)$
- D None of these

 [VIEW SOLUTION](#)

Question 59

What is the value of $ma(10, 4, le((la10, 5, 3), 5, 3))$?

- A 7
- B 6.5
- C 8
- D 7.5

 [VIEW SOLUTION](#)



Answers

1.12	2.D	3.C	4.B	5.4	6.8	7.A	8.A
9.8	10.16	11.B	12.B	13.3	14.A	15.B	16.16
17.C	18.A	19.D	20.C	21.397	22.A	23.2	24.D
25.A	26.47	27.B	28.C	29.5	30.C	31.C	32.A
33.C	34.B	35.D	36.A	37.D	38.A	39.B	40.C
41.A	42.C	43.D	44.C	45.B	46.11	47.2704	48.17
49.A	50.C	51.B	52.B	53.B	54.6	55.D	56.D
57.C	58.D	59.B					

Explanations

1.12

Let us break this up into 2 inequations [Let us assume x as m and y as n]

$$|1 + mn| < |m + n|$$

$$|m + n| < 5$$

Looking at these expressions, we can clearly tell that the graphs will be symmetrical about the origin.

Let us try out with the first quadrant and extend the results to the other quadrants.

We will also consider the +X and +Y axes along with the quadrant.

So, the first inequality becomes,

$$1 + mn < m + n$$

$$1 + mn - m - n < 0$$

$$1 - m + mn - n < 0$$

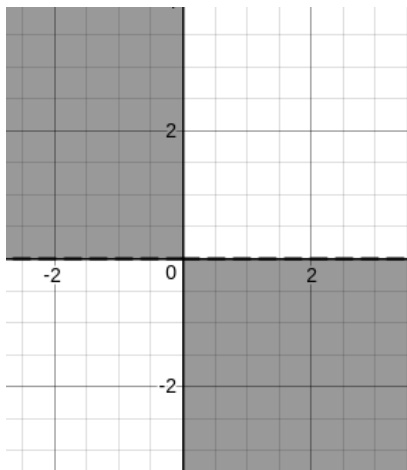
$$(1-m) + n(m-1) < 0$$

$$(1-m)(1-n) < 0$$

$$(m-1)(n-1) < 0$$

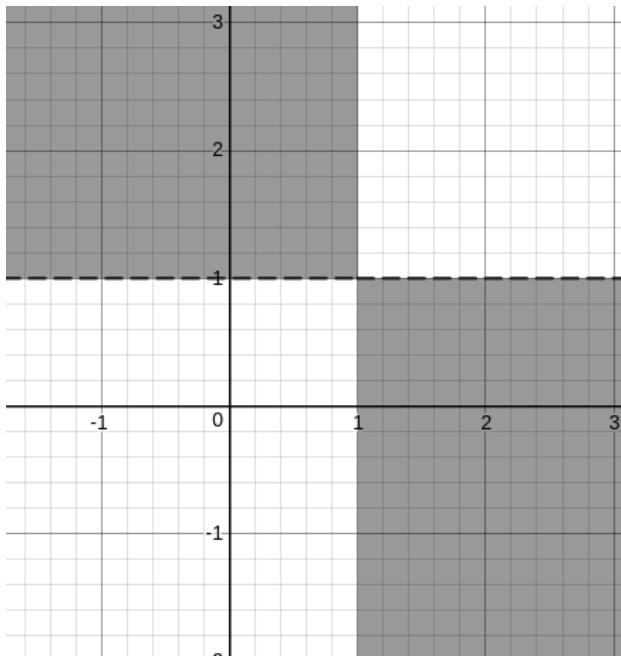
Let us try to plot the graph.

If we consider only $mn < 0$, then we get

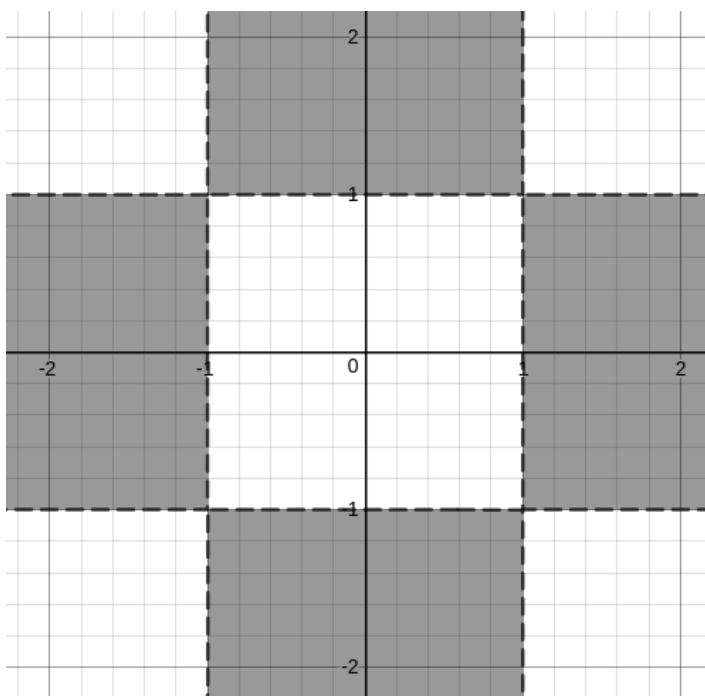


But, we have $(m-1)(n-1) < 0$, so we need to shift the graphs by one unit towards positive x and positive y.

So, we have,



But, we are only considering the first quadrant and the +X and +Y axes. Hence, if we extend, we get the following region.



So, if we look for only integer values, we get

(0,2), (0,3),.....

(0,-2), (0, -3),.....

(2,0), (3,0),

(-2,0), (-3,0),

Now, let us consider the other inequation as well, in which $|x + y| < 5$

Since one of the values is always zero, the modulus of the other value is less than or equal to 4.

Hence, we get

(0,2), (0,3), (0,4)

(0,-2), (0, -3), (0, -4)

(2,0), (3,0), (4,0)

(-2,0), (-3,0), (-4,0)

Hence, a total of 12 values.

 VIDEO SOLUTION

2. D

It is given that there are exactly 41 numbers, which can be expressed as the power of two, and exist between 8^m and 8^n , (where m, and n are positive integers, and $m < n$)

Hence, $2^{3m} < 41 \text{ numbers} < 2^{3n}$

Since, m is a positive integer, the least value of m is 1. Therefore, $2^{3m} = 2^3$, hence, the 41 numbers between them are $2^4, 2^5, 2^6, \dots, 2^{44}$.

Then the lowest possible value of 8^n is 2^{45} . Hence, the smallest value of n is $2^{45} = 8^n \Rightarrow 2^{3n} = 2^{45} \Rightarrow n = 15$

Hence, the smallest value of $m+n$ is $(15+1) = 16$

The correct option is D

 VIDEO SOLUTION

3. C

For the given expression value of x,y,z are distinct positive integers . So the value of expression will always be greater than value when all the 3 variables are equal . substitute $x=y=z$ we get minimum value of 6 .

$$(x^2(y+z) + y^2(x+z) + z^2(x+y))/xyz = x/z + x/y + y/z + y/x + z/y + z/x$$

Applying AM greater than or equal to GM, we get minimum sum = 6

 VIDEO SOLUTION

4. B

Taking lowest possible positive value of m i.e. 1 . Such that $a+b+c+d=5$, so atleast one of them must be greater than 1 ,

take $a=b=c=1$ and $d=2$

we get $a^2 + b^2 + c^2 + d^2 = 7$ which is equal to $4m^2 + 2m + 1$ for other values it is greater than $4m^2 + 2m + 1$. so option B

 VIEW SOLUTION

5. 4

Case 1: When $x^2 - 3x - 10 = 0$ and $x^2 - 10 \neq 0$

$$x^2 - 3x - 10 = 0 \text{ or, } (x-5)(x+2) = 0$$

or, $x = 5$ or -2

Case 2: $x^2 - 10 = 1$

$$x^2 - 11 = 0$$

No integer solutions

Case 3: $x^2 - 10 = -1$ and $x^2 - 3x - 10$ is even

$$x^2 - 9 = 0$$

or, $(x+3)(x-3)=0$

or, $x = -3$ and 3

for $x = -3$ and $+3$ $x^2 - 3x - 10$ is even

In total 4 values of x satisfy the equations

 VIDEO SOLUTION



6.8

The moduli will give out only non-negative outputs, and since we are to consider only integer values of x and y , this drastically reduces the possible cases.

We can get 2 from either $2+0$ or $1+1$

We get a $2+0$ form when either the first term or the second term is 0

The second term is 0; this is when $x=y$, in this case, $|2x|=2$, where x can be 1 or -1; therefore, the two cases are $(1,1)$ and $(-1,-1)$

The first term is 0; this is the case when $x = -y$, in this case, $|x - (-x)|=2$, giving $x=1$ or -1 yet again, here the two cases are $(1,-1)$ and $(-1,1)$

The other way we can get 2 is through $1+1$

This is possible when one of the terms is 0; if $y=0$, $|x|+|x|=2$, where x can be 1 or -1, giving two cases $(1,0)$ and $(-1,0)$

Similarly, for $x=0$, we get two cases, $(0,1)$ and $(0,-1)$

Therefore, there are 8 pairs of (x,y) that satisfy the given equation.

 VIDEO SOLUTION

7.A

For $n=3,4,5$ and $x \in [n, n+1)$ we have $\lfloor x \rfloor = n$, so the equation

$$\lfloor x^2 \rfloor = \lfloor x \rfloor^2 = n^2$$

$$x^2 \in [n^2, n^2 + 1), \text{ i.e. } x \in [n, \sqrt{n^2 + 1})$$

Thus for $3 \leq x \leq 6$

$$S = [3, \sqrt{10}) \cup [4, \sqrt{17}) \cup [5, \sqrt{26}) \cup 6$$

Option B and C have $\sqrt{10}$ included, which is not part of the original set. And Option D has $\sqrt{18}$. So, it is not possible.

Option A is the answer.

 VIDEO SOLUTION

8.A

$$y = \frac{4x-1}{17}$$

The integral values of x for which y is an integer are 13, 30, 47,.....

The values are in the form $17n + 13$, where $n \geq 0$

$$17n + 13 < 1000$$

$$\Rightarrow 17n < 987$$

$$\Rightarrow n < 58.05$$

$$\Rightarrow n \text{ can take values from 0 to 58} \Rightarrow \text{Number of values} = 59$$

VIDEO SOLUTION

9.8

We can see that at $n = 2$, $n^3 - 11n^2 + 32n - 28 = 0$ i.e. $(n-2)$ is a factor of $n^3 - 11n^2 + 32n - 28$

$$\frac{n^3 - 11n^2 + 32n - 28}{n - 2} = n^2 - 9n + 14$$

We can further factorize $n^2 - 9n + 14$ as $(n-2)(n-7)$.

$$n^3 - 11n^2 + 32n - 28 = (n - 2)^2(n - 7)$$

$$\Rightarrow n^3 - 11n^2 + 32n - 28 > 0$$

$$\Rightarrow (n - 2)^2(n - 7) > 0$$

Therefore, we can say that $n-7 > 0$

Hence, $n_{\min} = 8$

VIDEO SOLUTION

10.16

It is given, $x > y \geq 3$ and $x + y < 14$

Now for $y = 3$, the values of x can be 4,5,6,7,8,9,10 (7 cases)

Then for $y = 4$, the values of x can be 5,6,7,8,9 (5 cases)

Then for $y = 5$, the values of x can be 6,7,8 (3 cases)

Then for $y = 6$, the values of x will be 7 (1 case)

So, total number of cases = $1+3+5+7=16$ cases

VIDEO SOLUTION

A banner for 'CAT Mock Test' featuring two people on the left, the text 'CAT Mock Test' in large blue letters in the center, and 'Mocks designed exactly like the actual CAT' in a yellow box below it. On the right, there is a small icon of a clock and a book with the word 'MOCK' on it, and the 'cracku' logo above it.

11.B

$$x * x < y * y$$

$$\text{or } x + 0.5x - x^2 < y + 0.5y - y^2$$

$$y^2 - x^2 + 1.5x - 1.5y < 0$$

$$(y - x)(y + x) - 1.5(y - x) < 0$$

$$(y - x)(y + x - 1.5) < 0$$

$$(x - y)(1.5 - (x + y)) < 0$$

Now there will be two possibilities

$$x < y \text{ and } (x + y) < 1.5 \dots\dots\dots(i)$$

$$\text{or } x > y \text{ and } (x + y) > 1.5 \dots\dots\dots(ii)$$

Among all options only option B satisfies (ii).

Hence, option B is the correct answer.

VIDEO SOLUTION

12.B

$$\text{Given, } x^2 + (x - 2y - 1)^2 = -4y(x + y)$$

$$\Rightarrow x^2 + 4xy + 4y^2 + (x - 2y - 1)^2 = 0$$

$$\Rightarrow (x + 2y)^2 + (x - 2y - 1)^2 = 0$$

For the L.H.S. of the equation to be 0, each of the square terms should be 0 (as squares cannot be negative)

$$\Rightarrow x - 2y - 1 = 0 \Rightarrow x - 2y = 1$$

 VIDEO SOLUTION

13.3

Let us consider 3 cases:

1) $x = 0$, This is a solution, as both L.H.S and R.H.S will be equal (0) when $x = 0$. (1 solution)

2) $x > 0$

$$\Rightarrow 2x(x^2 + 1) = 5x^2$$

$$\Rightarrow 2(x^2 + 1) = 5x$$

$$\Rightarrow 2x^2 - 5x + 2 = 0 \Rightarrow 2x^2 - 4x - x + 2 = 0$$

$$\Rightarrow 2x(x - 2) - 1(x - 2) = 0$$

$$\Rightarrow (x - 2)(2x - 1) = 0 \Rightarrow x = 2 \text{ or } 1/2 \Rightarrow (1 \text{ integer solution})$$

3) $x < 0$

$$\Rightarrow -2x(x^2 + 1) = 5x^2$$

$$\Rightarrow 2x^2 + 5x + 2 = 0$$

$$\Rightarrow 2x^2 + 4x + x + 2 = 0$$

$$\Rightarrow 2x(x + 2) + 1(x + 2) = 0$$

$$\Rightarrow (x + 2)(2x + 1) = 0 \Rightarrow x = -2 \text{ or } -1/2 \Rightarrow (1 \text{ integer solution})$$

So, the total number of integer solutions are 0, 2, -2 $\Rightarrow$ 3.

 VIDEO SOLUTION

14.A

Given that the average marks of 4 girls and 6 boys is 24.

Let us assume 'b' is the marks scored by a boy and 'g' is the marks scored by a girl.

$$\Rightarrow 4g + 6b = 10 \times 24 = 240 \text{ ---(1)}$$

Given that, $b \leq g \leq 2b$

We need to find the distinct possible values of $2g + 6b = 2g + 240 - 4g = 240 - 2g$.

From (1)

$$\text{when } b = g \Rightarrow 10g = 240 \Rightarrow g = 24$$

$$\text{when } b = g/2 \Rightarrow 7g = 240 \Rightarrow g = 240/7$$

$$\Rightarrow 240 - 2g \text{ varies from } 240 - 2 \times 24 \text{ to } 240 - 2 \times 240/7$$

$$\Rightarrow 171.42 \text{ to } 192 \Rightarrow \text{Integer values of } 172 \text{ to } 192 \Rightarrow 21 \text{ values.}$$

 VIDEO SOLUTION

15.B

It is given that $\frac{x}{y} < \frac{x+3}{y-3}$, which can be written as $\frac{x}{y} - \frac{x+3}{y-3} < 0$

$$\Rightarrow \frac{x(y-3)-y(x+3)}{y(y-3)} < 0$$

$$\Rightarrow \frac{xy-3x-xy-3y}{y(y-3)} < 0$$

$$\Rightarrow \frac{-3(x+y)}{y(y-3)} < 0$$

$$\Rightarrow \frac{3(x+y)}{y(y-3)} > 0$$

From this inequality, we can say that, when $y < 0 \Rightarrow y(y-3) > 0$. Now to satisfy the given equation

$$\frac{3(x+y)}{y(y-3)} > 0,$$

$(x+y)$ must be greater than zero Hence, $x > 0$ and $|x| > |y|$

Therefore, the magnitude of x is greater than the magnitude of y .

Hence, $x > y$, and $|x| > |y| \Rightarrow -x < y$ (Since the magnitude of x is greater than the magnitude of y .)

The correct option is B.

[VIDEO SOLUTION](#)



16.16

It is given that if a certain amount of money is divided equally among n persons, each one receives Rs 352.

Hence, the total amount of money is $(352 \times n) = 352n$... Eq(1)

It is also known that if two persons receive Rs 506 each and the remaining amount is divided equally among the other persons, each of them receives less than or equal to Rs 330

Hence, the maximum amount of money with them $= 506 \times 2 + (n-2) \times 330 = 1012 + 330n - 660 = 352 + 330n$

Now, $352 + 330n \geq 352n$

$$\Rightarrow 22n \leq 352$$

$$\Rightarrow n \leq 16$$

Hence, the maximum value is 16

[VIDEO SOLUTION](#)

17.C

Given that $2pq - 20 = 52 - 2pq \Rightarrow 4pq = 72 \Rightarrow pq = 18$ ----(1)

$$\text{Now, } p^2 + q^2 - 29 = 2pq - 20 \Rightarrow p^2 + q^2 - 2pq = 9 \Rightarrow (p - q)^2 = 9 \Rightarrow p - q = \pm 3$$

$$\text{Also, } p^2 + q^2 - 29 = 2pq - 20 \Rightarrow p^2 + q^2 = 2pq + 9 = 2(18) + 9 = 45$$

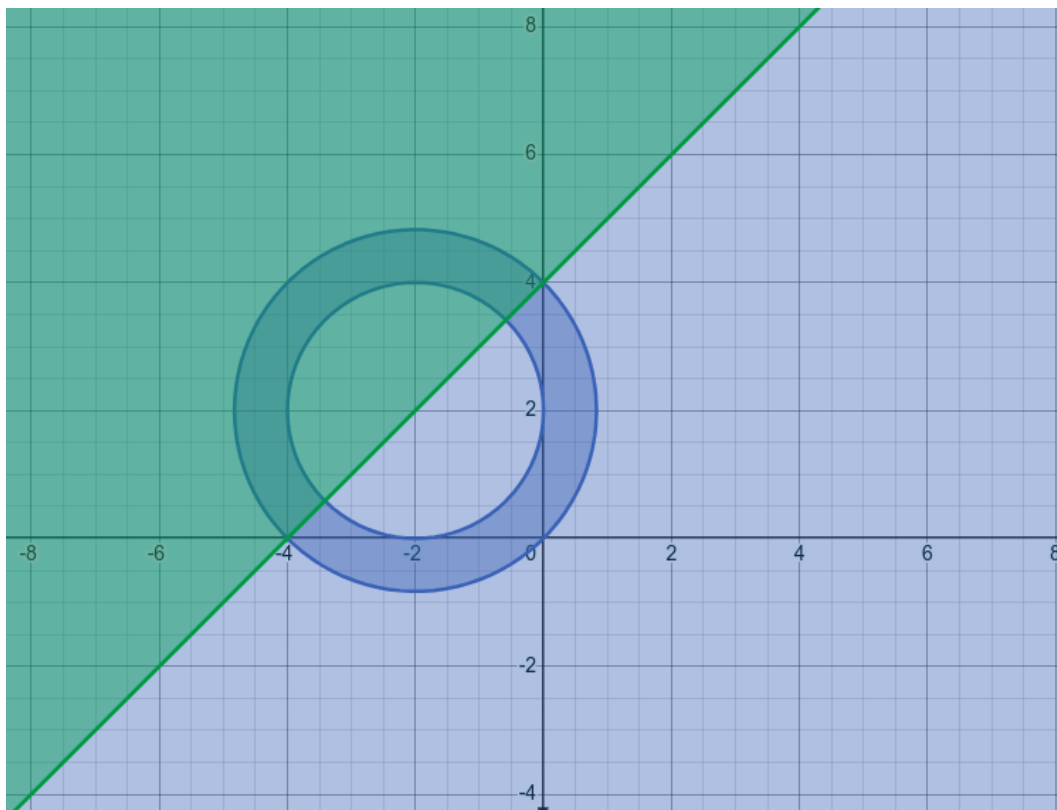
$$\text{Now, } p^3 - q^3 = (p - q)(p^2 + pq + q^2) = (p - q)(45 + 18) = (p - q)(63)$$

$\Rightarrow$ When $p - q = -3 \Rightarrow$ The value is $63(-3) = -189$ and when $p - q = 3 \Rightarrow$ The value is $63(3) = 189$.

$\Rightarrow$ The difference $= 189 - (-189) = 378$.

[VIDEO SOLUTION](#)

18.A



We have two inequalities,

$$y \geq x + 4$$

$$-4 \leq x^2 + y^2 + 4(x - y) \leq 0$$

The second inequality can be written as two separate inequalities,

$$-4 \leq x^2 + y^2 + 4(x - y) \text{ and } x^2 + y^2 + 4(x - y) \leq 0$$

The first inequality can be written as,

$$x^2 + y^2 + 4x - 4y + 4 \geq 0$$

$$x^2 + 4x + 4 + y^2 - 4y + 4 - 4 \geq 0$$

$$(x + 2)^2 + (y - 2)^2 \geq 4$$

The second inequality can be written as,

$$x^2 + y^2 + 4x - 4y \leq 0$$

$$x^2 + 4x + 4 + y^2 - 4y + 4 - 8 \leq 0$$

$$(x + 2)^2 + (y - 2)^2 \leq 8$$

Representing all three inequalities in the graph, we get the graph as shown above, and we must calculate the intersection of all three inequalities,

We can see that the line passes through the centre of both circles $(-2, 2)$, and the area obtained from the second inequality is the area between the two circles. So, the area of intersection of all three graphs is half of the area between both the circles as the line divides the circle in half, and we must only consider the area above the line as per the given inequality. We know that the area of the bigger circle is $\sqrt{8}$ and the area of the smaller circle is $\sqrt{4}$ from the equations of the circles as we know that equation of circle as $(x - a)^2 + (y - b)^2 = (\text{radius})^2$ where (a, b) is the centre of the circle.

The area of intersection

$$= \frac{1}{2} (\text{Area of bigger circle} - \text{Area of smaller circle})$$

$$\begin{aligned}
 &= \frac{1}{2} \left(\pi (\sqrt{8})^2 - \pi (\sqrt{4})^2 \right) \\
 &= \frac{1}{2} (8\pi - 4\pi) \\
 &= \frac{1}{2} (4\pi) \\
 &= 2\pi
 \end{aligned}$$

Therefore, the correct answer is option A.

[▶ VIDEO SOLUTION](#)

19. **D**

Substitute $x=6$ and $y=-6$,

$$x+4y = -18$$

$$x = 6, -4y = 24$$

$$-4x = -24, 5y = -30$$

So none of the options out of a,b or c satisfies .

[▶ VIDEO SOLUTION](#)

20. **C**

xy^2 will have it's least value when $y=-7$ and $x=2$ and equals 98.

$$\text{So } xy^2 > 98$$

x^2y will have it's least value when $y=-8$ and $x=3$ and equals -72.

$$\text{So, } x^2y > -72$$

$5xy$ will have it's least value when $y=-8$ and $x=3$ and equals -120

$$\text{So, } 5xy > -120$$

So, of the three expressions, the least possible value is that of $5xy$

[▶ VIEW SOLUTION](#)



21. **397**

We have $p + q + r = 900$, and r is a perfect square lying between 150 and 500.

Let q be a non-variable number that does not have a range.

$$\text{We have } 0.3q \leq p \leq 0.7q, \text{ or } q + 0.3q \leq p + q \leq q + 0.7q \text{ or } 1.3q \leq p + q \leq 1.7q$$

$$\text{Also, we can rewrite } p + q = 900 - r$$

$$\text{Therefore, } 1.3q \leq 900 - r \leq 1.7q$$

Since the extreme values (maximum and minimum) of p depend on the value of q ($0.3q$ and $0.7q$), we can maximise or minimise r to find the extreme values of p .

Since $1.3q \leq 900 - r$, the minimum possible value of q would be when $1.3q = 900 - r$ and when r is maximum. The maximum value of r is 484. We get, $1.3q = 900 - 484$, or $q = 320$.

This gives the minimum possible value of p as $0.3 \times 320 = 96$.

Since $900 - r \leq 1.7q$, the maximum possible value of q would be when $900 - r = 1.7q$ and when r is minimum. The minimum value of r is 169. We get $1.7q = 900 - 169$ or $q = 430$

This gives the maximum possible value of p as $0.7 \times 430 = 301$

The sum of the maximum and minimum values of p is $96 + 301 = 397$.

 [VIEW SOLUTION](#)

22. A

We are asked to solve

$$x^2 - |x + 9| + x > 0$$

Split into two cases based on the absolute value.

Case 1: $x + 9 \geq 0 \Rightarrow x \geq -9$

$$|x + 9| = x + 9$$

Inequality becomes: $x^2 - (x + 9) + x > 0 \Rightarrow x^2 - 9 > 0 \Rightarrow (x - 3)(x + 3) > 0$

So $x < -3$ or $x > 3$. Combined with $x \geq -9$, we get $x \in [-9, -3) \cup (3, \infty)$

Case 2: $x + 9 < 0 \Rightarrow x < -9$

$$|x + 9| = -(x + 9) = -x - 9$$

Inequality becomes: $x^2 - (-x - 9) + x > 0 \Rightarrow x^2 + 2x + 9 > 0$

The quadratic $x^2 + 2x + 9$ has discriminant $(4 - 36 = -32 < 0)$, so always positive. But in this case ($x < -9$), so inequality is satisfied. Thus ($x < -9$) also works.

$$x < -3 \text{ or } x > 3$$

So the solution set is $(-\infty, -3) \cup (3, \infty)$

 [VIDEO SOLUTION](#)

23. 2

Given expression: $(a - b)^2 + (a - c)^2 + (a - d)^2$

The given expression has just the sum of squares of the terms. So, the minimum value is either zero or positive.

If we can make all the values equal, we can get zero. But since all the values are integers and the sum 46 is not divisible by 4, we can't make everything equal.

So, the nearest four values are 12, 11, 11, 12.

With this, the minimum value is $(12 - 11)^2 + (12 - 11)^2 + (12 - 12)^2 = 2$

 [VIDEO SOLUTION](#)

24. D

Since the square root can be positive or negative we will get two cases for each of the equation.

For the first one,

$$a + 3 = 3b \dots i$$

$$a + 3 = -3b \dots ii$$

For the second one,

$$a - 1 = 2(b - 1) \dots iii$$

$$a - 1 = 2(1 - b) \dots \text{iv}$$

we have to solve i and iii, i and iv, ii and iii, ii and iv.

Solving i and iii,

$a + 3 = 3b$ and $a = 2b - 1$, solving, we get $a = 3$ and $b = 2$, which is not what we want.

Solving i and iv

$a + 3 = 3b$ and $a = 3 - 2b$, solving, we get $b = 1.2$, which is not possible.

Solving ii and iii

$a + 3 = -3b$ and $a = 2b - 1$, solving, we get $b = 0.4$, which is not possible.

Solving ii and iv,

$a + 3 = -3b$ and $a = 3 - 2b$, solving, we get $a = 15$ and $b = -6$ which is what we want.

$$\text{Thus, } \frac{a^2}{b^2} = \frac{25}{4}$$

 VIDEO SOLUTION

25. A

$$f(x) = \frac{x^2 + 2x - 15}{x^2 - 7x - 18} < 0$$

$$\frac{(x+5)(x-3)}{(x-9)(x+2)} < 0$$

We have four inflection points -5, -2, 3, and 9.

For $x < -5$, all four terms $(x+5)$, $(x-3)$, $(x-9)$, $(x+2)$ will be negative. Hence, the overall expression will be positive.

Similarly, when $x > 9$, all four terms will be positive.

When x belongs to $(-2, 3)$, two terms are negative and two are positive. Hence, the overall expression is positive again.

We are left with the range $(-5, -2)$ and $(3, 9)$ where the expression will be negative.

 VIDEO SOLUTION



CAT Syllabus PDF



26. 47

$$\text{Let } \log_2 n = y$$

$$\frac{4-y}{3-\frac{y}{2}} < 0$$

$$(4 - y) \left(3 - \frac{y}{2}\right) < 0$$

$$(4 - y)(6 - y) < 0$$

$$(y - 4)(y - 6) < 0$$

$$4 < y < 6$$

$$4 < \log_2 n < 6$$

$$2^4 < n < 2^6$$

$$16 < n < 64$$

n can take values from 17 to 63(inclusive).

The number of n values possible = 47

 VIDEO SOLUTION

27. **B**

$$y = 38 \Rightarrow x = 1$$

$$y = 36 \Rightarrow x = 2$$

...

...

$$y = 14 \Rightarrow x = 13$$

$$y = 12 \Rightarrow x = 14 \Rightarrow \text{Cases from here are not valid as } x > y.$$

Hence, there are 13 solutions.

 VIDEO SOLUTION

28. **C**

The equation is not satisfied for only $x = 2y$.

Using statements B and C, i.e., $x = 2z$ and $2x = z$, we see that the equation is not satisfied.

Using statements A and B, i.e., $x = 2y$ and $x = 2z$, i.e., $z = y = x/2$, the equation is satisfied.

Option c) is the correct answer.

 VIEW SOLUTION

29. **5**

It is given that $5^{n-1} < 3^{n+1}$, where n is a natural number. By inspection, we can say that the inequality holds when $n = 1, 2, 3, 4$, and 5 .

Now, we need to find the least integer value of m that satisfies $3^{n+1} < 2^{n+m}$

For, $n = 1$, the least integer value of m is 3.

For, $n = 2$, the least integer value of m is 3

For, $n = 3$, the least integer value of m is 4.

For, $n = 4$, the least integer value of m is 4.

For, $n = 5$, the least integer value of m is 5.

Hence, the least integer value of m such that for all the values of n , the equation holds is 5.3

 VIDEO SOLUTION

30. **C**

It is given that the population of the town in 2020 was 100000. The population decreased by $y\%$ from the year 2020 to 2021 and increased by $x\%$ from the year 2021 to 2022, where x and y are two natural numbers.

Hence, the population in 2021 is $100000 \left(\frac{100-y}{100} \right)$.

The population in 2022 is $100000 \left(\frac{100-y}{100} \right) \left(\frac{100+x}{100} \right)$

It is also given that the population in 2022 was greater than the population in 2020 and the difference between x and y is 10.

Hence,

$$100000 \left(\frac{100-y}{100} \right) \left(\frac{100+x}{100} \right) > 100000, \text{ and } (x-y) = 10$$

$$\Rightarrow 100000 \left(\frac{100-y}{100} \right) \left(\frac{110+y}{100} \right) > 100000$$

$$\Rightarrow \frac{100-y}{100} \left(\frac{110+y}{100} \right) > 1$$

To get the minimum possible value of 2021, we need to increase the value of y as much as possible.

$$\text{Hence, } (100 - y) \{(100 + y) + 10\} > 10000$$

$$\Rightarrow 10000 - y^2 + 1000 - 10y > 10000$$

$$\Rightarrow y^2 + 10y < 1000$$

$$\Rightarrow y^2 + 10y + 25 < 1025$$

$$\Rightarrow (y + 5)^2 = 1024 < 1025$$

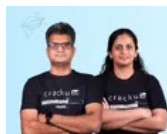
$$\Rightarrow (y + 5)^2 = 32^2$$

$$\Rightarrow y = 27$$

Hence, the population in 2021 is $100000 \times (100 - 27) = 73000$

The correct option is C

 VIDEO SOLUTION



CAT Formula PDF



31. **C**

Let $p = q = r = 1$

So, the value of the expression becomes $1/3 + 1/3 + 1/3 = 1$

If we substitute these values, options a), b) and d) do not satisfy.

Option c) is the answer.

 VIDEO SOLUTION

32. **A**

After solving given equation, we will have inequality resolved to:

$$(x-1)(x-2) > 0$$

Or we can say range of x will be as follows:

$$x < 1; x > 2$$

Hence, option A has a set of values which don't lie in the possible range of x.

So the answer will be A.

 VIDEO SOLUTION

33. **C**

$$\text{Case 1 : } x \geq \frac{40}{3}$$

$$\text{we get } 3x - 20 + 3x - 40 = 20$$

$$6x = 80$$

$$x = \frac{80}{6} = \frac{40}{3} = 13.33$$

$$\text{Case 2 : } \frac{20}{3} \leq x < \frac{40}{3}$$

$$\text{we get } 3x - 20 + 40 - 3x = 20$$

we get $20=20$

So we get $x \in \left[\frac{20}{3}, \frac{40}{3} \right]$

Case 3 $x < \frac{20}{3}$

we get $20-3x+40-3x=20$

$40=6x$

$x=\frac{20}{3}$

but this is not possible

so we get from case 1,2 and 3

$\frac{20}{3} \leq x \leq \frac{40}{3}$

Now looking at options

we can say only option C satisfies for all x .

Hence $7 < x < 12$.

[VIDEO SOLUTION](#)

34. B

We know $w = vz/u$ so taking max value of u and min value of v and z to get min value of w which is -4 .

Similarly taking min value of u and max value of v and z to get max value of w which is 4

Take $v = 1, z = -2$ and $u = -0.5$, we get $w = 4$

Take $v = -1, z = -2$ and $u = -0.5$, we get $w = -4$

[VIDEO SOLUTION](#)

35. D

$$n^3 - 7n^2 + 11n - 5 = (n-1)(n^2 - 6n + 5) = (n-1)(n-1)(n-5)$$

This is positive for $n > 5$

So, $m = 6$

[VIDEO SOLUTION](#)



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36. A

$\sqrt{\log_e \frac{4x-x^2}{3}}$ will be real if $\log_e \frac{4x-x^2}{3} \geq 0$

$$\frac{4x-x^2}{3} \geq 1$$

$$4x - x^2 - 3 \geq 0$$

$$x^2 - 4x + 3 \leq 0$$

$$1 \leq x \leq 3$$

[VIDEO SOLUTION](#)

37. D

$$|r-6| = 11 \Rightarrow r = -5 \text{ or } 17$$

$$|2q-12| = 8 \Rightarrow q = 10 \text{ or } 2$$

So, the minimum possible value of $q/r = 10/(-5) = -2$

VIDEO SOLUTION

38. A

Try to solve this type of questions using the options.

Substitute 0 first $\Rightarrow$ We get $-2 \leq 0$, which is correct. Hence, 0 must be in the solution set.

Substitute 8 $\Rightarrow 4 + 2 - 2 \leq 0 \Rightarrow 6 \leq 0$, which is false. Hence, 8 must not be in the solution set.

$\Rightarrow$ Option 1 is the answer.

VIDEO SOLUTION

39. B

This kind of questions must be solved using the counter example method.

$x = 100$ and $y = -1/2$ rules out option a)

$x = 3$ and $y = 0$ rules out options c) and d)

Option b) is correct.

VIDEO SOLUTION

40. C

The given equations are $x + y + z = 5$ -- (1), $xy + yz + zx = 3$ -- (2)

$$xy + yz + zx = 3$$

$$x(y + z) + yz = 3$$

$$\Rightarrow x(5 - x) + y(5 - x - y) = 3$$

$$\Rightarrow -y^2 - y(5 - x) - x^2 + 5x = 3$$

$$\Rightarrow y^2 + y(x - 5) + (x^2 - 5x + 3) = 0$$

The above equation should have real roots for y , $\Rightarrow$ Determinant ≥ 0

$$\Rightarrow b^2 - 4ac \geq 0$$

$$\Rightarrow (x - 5)^2 - 4(x^2 - 5x + 3) \geq 0$$

$$\Rightarrow 3x^2 - 10x - 13 \leq 0$$

$$\Rightarrow -1 \leq x \leq \frac{13}{3}$$

Hence maximum value x can take is $\frac{13}{3}$, and the corresponding values for y, z are $\frac{1}{3}, \frac{1}{3}$

VIDEO SOLUTION

CAT Percentile Predictor



41. A

The expressions in the four options can be expanded as

$$xyz - yz; xyz - xz; xyz - xy \text{ and } xyz + xz$$

The closest value to xyz would be $xyz - yz$, as yz is the least value among yz, xz and xy .

Option a) is the correct answer.

VIDEO SOLUTION

42. C

Approach 1:

The given expression is symmetric in x and y and the limiting condition (x+y=1) is also symmetric in x and y.

=> This means that the expression attains the minimum value when x = y

$$x=y=1/2$$

$$\text{So, the value} = \left(x + \frac{1}{x}\right)^2 + \left(y + \frac{1}{y}\right)^2 = \left(2 + \frac{1}{2}\right)^2 + \left(2 + \frac{1}{2}\right)^2 = 12.5$$

Approach 2:

$$(x + 1/x)^2 + (y + 1/y)^2 = (x + 1/x + y + 1/y)^2 - 2 * (x + 1/x)(y + 1/y)$$

Let $x+1/x$ and $y+1/y$ be two terms. Thus $(x+1/x+y+1/y)/2$ would be their Arithmetic Mean (AM) and $\sqrt{(x + 1/x)(y + 1/y)}$ would be their Geometric Mean (GM).

Therefore, we can express the above equation as $(x + 1/x)^2 + (y + 1/y)^2 = 4AM^2 - GM^2$. As $AM \geq GM$, the minimum value of expression would be attained when $AM = GM$.

When $AM = GM$, both terms are equal. That is $x+1/x = y+1/y$.

Substituting $y=1-x$ we get

$$x+1/x = (1-x) + 1/(1-x)$$

$$\text{On solving we get } 2x-1 = (2x-1)/x(1-x)$$

$$\text{So either } 2x-1 = 0 \text{ or } x(1-x) = 1$$

$$x(1-x) = x * y$$

As x and y are positive numbers whose sum = 1, $0 < x, y < 1$. Hence, their product cannot be 1.

$$\text{Thus, } 2x-1 = 0 \text{ or } x=1/2$$

$$\Rightarrow y = 1/2$$

$$\text{So, the value} = \left(x + \frac{1}{x}\right)^2 + \left(y + \frac{1}{y}\right)^2 = \left(2 + \frac{1}{2}\right)^2 + \left(2 + \frac{1}{2}\right)^2 = 12.5$$

 VIDEO SOLUTION

43. D

$$\text{Now } \frac{x}{\sqrt{1+x^4}} = \frac{1}{\sqrt{\frac{1+x^4}{x^2}}} = \frac{1}{\sqrt{\frac{1}{x^2} + x^2}}$$

Applying A.M $\geq$ G.M.

$$\frac{\left(\frac{1}{x^2} + x^2\right)}{2} \geq 1 \text{ or } \frac{1}{x^2} + x^2 \geq 2 \text{ Substituting we get the maximum possible value of the equation as } \frac{1}{\sqrt{2}}$$

 VIDEO SOLUTION

44. C

Since the product is constant,

$$(a+b+c+d)/4 \geq (abcd)^{1/4}$$

We know that $abcd = 1$.

Therefore, $a+b+c+d \geq 4$

$$(a+1)(b+1)(c+1)(d+1)$$

$$= 1 + a + b + c + d + ab + ac + ad + bc + bd + cd + abc + bcd + cda + dab + abcd$$

We know that $abcd = 1$

$$\text{Therefore, } a = 1/bcd, b = 1/acd, c = 1/bda \text{ and } d = 1/abc$$

$$\text{Also, } cd = 1/ab, bd = 1/ac, bc = 1/ad$$

The expression can be clubbed together as $1 + abcd + (a + 1/a) + (b + 1/b) + (c + 1/c) + (d +$

$$1/d) + (ab + 1/ab) + (ac + 1/ac) + (ad + 1/ad)$$

For any positive real number x , $x + 1/x \geq 2$

Therefore, the least value that $(a + 1/a), (b + 1/b), \dots, (ad + 1/ad)$ can take is 2.

$$(a + 1)(b + 1)(c + 1)(d + 1) \geq 1 + 1 + 2 + 2 + 2 + 2 + 2 + 2 + 2 + 2$$

$$\Rightarrow (a + 1)(b + 1)(c + 1)(d + 1) \geq 16$$

The least value that the given expression can take is 16. Therefore, option C is the right answer.

[VIDEO SOLUTION](#)

45. B

$$x_0 = \max(x_1, x_2, \dots, x_{12})$$

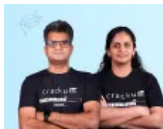
x_0 will be minimum if $x_1, x_2, \dots, x_{12}$ are close to each other

$$100/12 = 8.33$$

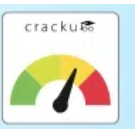
$\therefore \max(x_1, x_2, \dots, x_{12})$ will be minimum if $(x_1, x_2, \dots, x_{12}) = (9, 9, 9, 9, 8, 8, 8, 8, 8, 8, 8, 8)$

$\therefore$ Option B is correct.

[VIDEO SOLUTION](#)



CAT Score Calculator



46. 11

$$(n - 5)(n - 10) - 3(n - 2) \leq 0$$

$$\Rightarrow n^2 - 15n + 50 - 3n + 6 \leq 0$$

$$\Rightarrow n^2 - 18n + 56 \leq 0$$

$$\Rightarrow (n - 4)(n - 14) \leq 0$$

$\Rightarrow$ Thus, n can take values from 4 to 14. Hence, the required number of values are $14 - 4 + 1 = 11$.

[VIDEO SOLUTION](#)

47. 2704

$$\text{Now we have } 2601 \left(1 + \frac{1}{x}\right) \left(1 + \frac{1}{y}\right) = 2601 \left(\frac{xy+y+x+1}{xy}\right)$$

Now we know that $x+y=102$. Substituting it in the above equation

$$2601 \left(\frac{xy+y+x+1}{xy}\right) = 2601 \left(\frac{103}{xy} + 1\right)$$

Maximum value of xy can be found out by $AM \geq GM$ relationship

$$\frac{x+y}{2} \geq \sqrt{xy} \text{ or } \sqrt{xy} \leq 51 \text{ or } xy \leq 2601$$

Hence the maximum value of " xy " is 2601. Substituting in the above equation we get

$$2601 \left(\frac{103+2601}{2601}\right) = 2704$$

[VIDEO SOLUTION](#)

48. 17

$$\text{We have } 2x + 5y = 99 \text{ or } x = \frac{(99-5y)}{2}$$

$$\text{Now } x \geq y \geq -20; \text{ So } \frac{(99-5y)}{2} \geq y; 99 \geq 7y \text{ or } y \leq \approx 14$$

So $-20 \leq y \leq 14$. Now for this range of "y", we have to find all the integral values of "x". As the coefficient of "x" is 2,

then $(99 - 5y)$ must be even, which will happen when "y" is odd. However, there are only 17 odd values of "y" between -20 and 14.

Hence the number of possible values is 17.

 VIDEO SOLUTION

49. A

There are two critical points for the inequality to consider: $x = -5$ and $x = 3/2$



Region I: x is greater than $3/2$

In this scenario, both the terms would be positive; cross-multiplying, we get the relation $2x - 3 \leq x + 5$

Giving the boundary $x \leq 8$, hence giving us the valid range as $\frac{3}{2} < x \leq 8$

Region II: $-5 < x < \frac{3}{2}$

In this case, the right-hand side will be a negative value, and hence, the sign would change when multiplying, giving the inequality

$$2x - 3 \geq x + 5$$

Which will give $x > 8$, which is out of bounds for this region

Another way is to put a value in the region to check for the validity of the inequality; by putting $x = 0$, we could see that the inequality does not hold in this region

Region III: x less than -5

In this scenario, both the terms are negative, essentially giving us the same boundary as region 1; we take the lower bounds, giving us that x has to be less than 5

Therefore, for the given inequality to hold true $x < -5$ or $\frac{3}{2} < x \leq 8$

Hence, Option A is the correct answer.

 VIDEO SOLUTION

50. C

Let the roots of the equation $x^2 + (a + 3)x - (a + 5) = 0$ be equal to p, q

Hence, $p + q = -(a + 3)$ and $p \times q = -(a + 5)$

$$\text{Therefore, } p^2 + q^2 = a^2 + 6a + 9 + 2a + 10 = a^2 + 8a + 19 = (a + 4)^2 + 3$$

As $(a + 4)^2$ is always non negative, the least value of the sum of squares is 3

 VIDEO SOLUTION



A banner for 'CAT Crash Course' featuring two people on the left and a 'cracku' logo on the right. The text 'CAT Crash Course' is in large blue letters, and 'Live Classes by IIMA alumni' is in a red box below it.

51. B

$-n+2 \geq 0$
 or $n \leq 2$
 and $2n \geq 4$
 or $n \geq 2$

So we can take only one value of n i.e. 2

 VIDEO SOLUTION

52. B

We can consider the quadrants of a graph:

First quadrant: Both x and y are positive

This would change the equation to $2x+y=15$ and $x=20$, giving a negative value of y ; hence, this is not the case.

Second quadrant: x is negative, but y is positive

This would change the equations to $y=15$ and $x=20$, giving a positive value of x , which hence can not be the case.

Third quadrant: Both x and y are negative

This would change the equation to $y=15$ and $x-2y=20$; this gives a positive value of y and hence can not be the case.

Fourth quadrant: x is positive, but y is negative

This would change the equations to $2x+y=15$ and $x-2y=20$; this gives the value of x as 10 and y as -5, which would lie in the fourth quadrant.

The value of $x-y$ would be $10-(-5)=15$

Therefore, Option B is the correct answer.

 VIDEO SOLUTION

53. B

We have $|n - 60| < |n - 100| < |n - 20|$.

Now, the difference inside the modulus signified the distance of n from 60, 100, and 20 on the number line. This means that when the absolute difference from a number is larger, n would be further away from that number.

Example: The absolute difference of n and 60 is less than that of the absolute difference between n and 20. Hence, n cannot be ≤ 40 , as then it would be closer to 20 than 60, and closer on the number line would indicate lesser value of absolute difference. Thus we have the condition that $n > 40$.

The absolute difference of n and 100 is less than that of the absolute difference between n and 20. Hence, n cannot be ≤ 60 , as then it would be closer to 20 than 100. Thus we have the condition that $n > 60$.

The absolute difference of n and 60 is less than that of the absolute difference between n and 100. Hence, n cannot be ≥ 80 , as then it would be closer to 100 than 60. Thus we have the condition that $n < 80$.

The number which satisfies the conditions are 61, 62, 63, 64.....79. Thus, a total of 19 numbers.

Alternatively

as per the given condition : $|n - 60| < |n - 100| < |n - 20|$.

Dividing the range of n into 4 segments. ($n < 20$, $20 < n < 60$, $60 < n < 100$, $n > 100$)

1) For $n < 20$.

$|n-20| = 20-n$, $|n-60| = 60-n$, $|n-100| = 100-n$

considering the inequality part : $|n - 100| < |n - 20|$

$100 - n < 20 - n$,

No value of n satisfies this condition.

2) For $20 < n < 60$.

$$|n-20| = n-20, |n-60| = 60-n, |n-100| = 100-n.$$

$$60-n < 100-n \text{ and } 100-n < n-20$$

For $100-n < n-20$.

$$120 < 2n \text{ and } n > 60. \text{ But for the considered range } n \text{ is less than } 60.$$

3) For $60 < n < 100$

$$|n-20| = n-20, |n-60| = n-60, |n-100| = 100-n$$

$$n-60 < 100-n \text{ and } 100-n < n-20.$$

For the first part $2n < 160$ and for the second part $120 < 2n$.

n takes values from 6179.

A total of 19 values

4) For $n > 100$

$$|n-20| = n-20, |n-60| = n-60, |n-100| = n-100$$

$$n-60 < n-100.$$

No value of n in the given range satisfies the given inequality.

Hence a total of 19 values satisfy the inequality.

 VIDEO SOLUTION

54. 6

$$(\sqrt[7]{10})(\sqrt[7]{10})^2 \dots (\sqrt[7]{10})^n > 999$$

$$(\sqrt[7]{10})^{1+2+\dots+n} > 999$$

$$10^{\frac{1+2+\dots+n}{7}} > 999$$

For minimum value of n ,

$$\frac{1+2+\dots+n}{7} = 3$$

$$1+2+\dots+n = 21$$

We can see that if $n = 6$, $1+2+3+\dots+6 = 21$.

 VIDEO SOLUTION

55. D

Substitute value of $u = v = 2$, $w = 4$ and $m = 1$. Here the condition holds and options A and B are false. Hence, we can eliminate options A and B.

Substitute $u = v = 1$, $w = 2$ and $m = 1$. Here $m = \min(u, v, w)$. Hence, option C also does not hold. Hence, we can eliminate option C.

Option d) is the correct answer.

 VIEW SOLUTION



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56. D

Consider 3 family. Let 1st one have 2 A , 1B and 2 G , 2nd one have 2 A , 2B and 1 G, 3rd one have 2 A , 2B and 1 G . SO total A-6 , B - 5 , g - 4 , F - 3 . Hence minimum is 3 .

[▶ VIDEO SOLUTION](#)

57. **C**

Given expression can be reduced to

$$le(15, \min(10, 15-9), le(9, 8, 12))$$

$$\text{Or } le(15, 6, 1) = 9$$

[▶ VIEW SOLUTION](#)

58. **D**

Best approach to these type question remain assuming values and checking

$$\text{Case - 1. } x=8; y=7; z=5$$

$$la(x, y, z) = 12$$

$$le(x, y, z) = 2$$

$$ma(x, y, z) = 7$$

Case -2: Let us try to find values for which $la(x, y, z)$ and $le(x, y, z)$ would be equal. In such a case, $ma(x, y, z)$ would also be the same.

$$\text{So } \max(x-y, y-z) = \min(x+y, y+z)$$

$$\text{As } x > y > z > 0, \min(x+y, y+z) = y+z$$

$$\text{So } \max(x-y, y-z) = y+z$$

$$\text{Either } x-y=y+z \text{ or } y-z=y+z$$

$$\text{So } x=2y+z \text{ or } z=0$$

But z cannot be 0 according to given condition.

$$\text{So, } x=2y+z$$

$$\text{Let us assume } y=2 \text{ and } z=1$$

$$\text{So } x=5$$

$$la(x, y, z) = 3$$

$$le(x, y, z) = 3$$

$$ma(x, y, z) = 3$$

based on these two cases we can deduce that non of the given options holds true.

So the correct option to choose is D - None of these.

[▶ VIEW SOLUTION](#)

59. **B**

$$Ma(10, 4, le((la(10, 5, 3), 5, 3)))$$

$$\text{Or } Ma(10, 4, le(8, 5, 3))$$

$$\text{Or } Ma(10, 4, 3)$$

$$\text{Or } \frac{1}{2}(6 + 7) = 6.5$$

[▶ VIEW SOLUTION](#)